

Enzyme and Metabolism Study Guide

C1.1.1— Enzymes as catalysts. *Students should understand the benefit of increasing rates of reaction in cells.*

- Define catalyst.

An enzyme

A substance that increases the rate of chemical reaction without being chemically changed

- State the role of enzymes in the chemical reactions on which life is based.

Biological catalysts

Each step within a metabolic pathway has enzyme catalysis

- Lowering activation energy, specificity, regulation, reusability

- State that enzymes speed up chemical reactions without being altered, so they can be reused.

C1.1.2— Role of enzymes in metabolism. *Students should understand that metabolism is the complex network of interdependent and interacting chemical reactions occurring in living organisms. Because of enzyme specificity, many different enzymes are required by living organisms, and control over metabolism can be exerted through these enzymes.*

- Define metabolism.

Sum of all chemical reactions that occur within living organisms

- Define specificity in relation to enzyme structure and function.

The enzyme's structure is specific to the active site that it has and is specific to the substrate it binds to in order to facilitate biochemical reactions efficiently

- Outline how enzymes regulate control of metabolism.

Steps to make end product, each step has enzyme that helps reactions

C1.1.3— Anabolic and catabolic reactions. *Examples of anabolism should include the formation of macromolecules from monomers by condensation reactions including protein synthesis, glycogen formation and photosynthesis. Examples of catabolism should include hydrolysis of macromolecules into monomers in digestion and oxidation of substrates in respiration.*

- Contrast anabolic and catabolic reactions.

Anabolic: to build up

Catabolic: to break down

- List three examples of anabolic processes.

Formation of glycogen

Photosynthesis

Protein synthesis

- List three examples of catabolic processes.

Breakdown of glycogen

Cellular respiration

Lipolysis (breakdown of triglycerides into glycerol and fatty acids)

C1.1.4— Enzymes as globular proteins with an active site for catalysis. *Include that the active site is composed of a few amino acids only, but interactions between amino acids within the overall three-dimensional structure of the enzyme ensure that the active site has the necessary properties for catalysis.*

- Outline properties of globular proteins.

Shape: spherical

Role: functional (catalysis and transport)

Solubility in water: mostly

Sequence of amino acids: irregular amino acid sequence (complex folding)

Stability: more sensitive to change in Ph and temp

Examples: hemoglobin, insulin, catalase

- Explain the relationship between enzyme structure and enzyme specificity, including the structure and function of the active site.

Because enzymes have specific shapes, the active site is specific and only binds to substrates that fit

C1.1.5— Interactions between substrate and active site to allow induced-fit binding.

Students should recognize that both substrate and enzymes change shape when binding occurs.

- Outline the stages of enzyme catalysis of a chemical reaction.
 1. Collision: enzyme and substrate collide because they are in constant motion
 2. Catalysis: enzyme catalyzing reaction. Substrate binds to enzyme site
 3. Release: products leave active site. Enzyme is left unchanged, can be reused

- Describe the induced fit model of enzyme binding.

The substrate slightly changes shape for enzyme's active site

Both enzyme and substrate change shape, improving the binding and lowering activation energy

C1.1.6- Role of molecular motion and substrate-active site collisions in enzyme catalysis.

Movement is needed for a substrate molecule and an active site to come together. Sometimes large substrate molecules are immobilized while sometimes enzymes can be immobilized by being embedded in membranes.

- Explain the role of random collisions in the binding of the substrate with the enzyme active site.

Because they're in constant motion bc of molecular motion = random collisions

Random collisions: needed for the binding process

If more enzyme or substrate then higher chance of collision

- Compare enzyme and substrate movement involved in reactions that occur in the cytoplasm, with large substrates and with immobilized enzymes.

Many enzyme-catalyzed reactions happen in cytoplasm, usually substrate moves more bc smaller

If substrate large in size then they don't move much, meaning, enzyme has to move to substrate

Immobilized enzymes cant move since embedded in membranes, meaning, substrate has to move to enzyme

C1.1.7- Relationships between the structure of the active site, enzyme–substrate specificity and denaturation. *Students should be able to explain these relationships.*

- Discuss variation in specificity of different enzymes.

Enzyme specific to substrate, making only some allow to bind

Some not so specific and allow bindings to other substrates

- Define denaturation.

Change in the 3D structure of a protein that results in loss of function

- Outline the causes and effects of denaturation on enzyme structure and function.

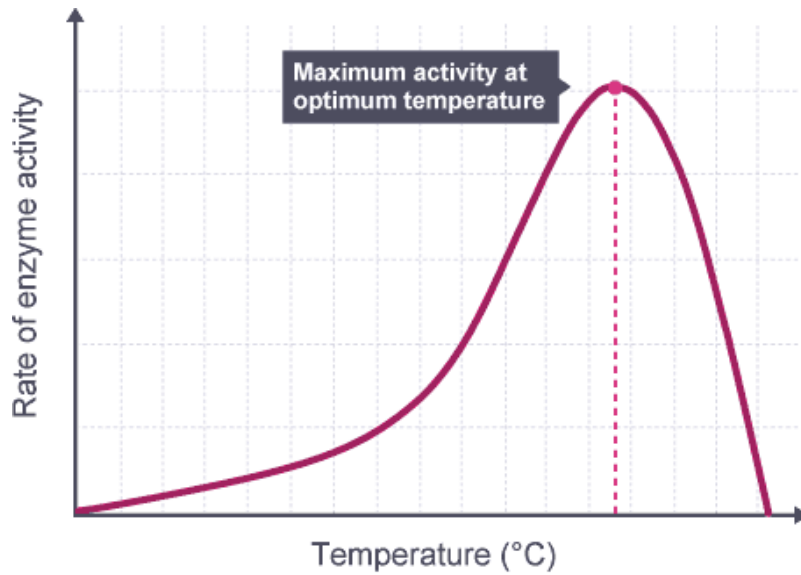
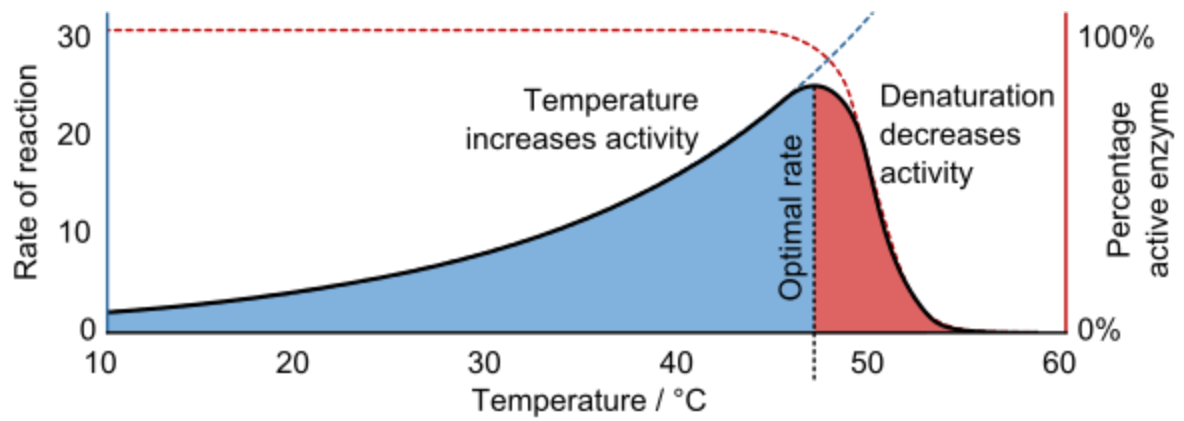
If enzyme structure changed, substrate no longer able to bind then won't be able to function (prevents binding, enzyme no longer able to catalyze reaction)

C1.1.8- Effects of temperature, pH and substrate concentration on the rate of enzyme activity. *The effects should be explained with reference to collision theory and denaturation.*

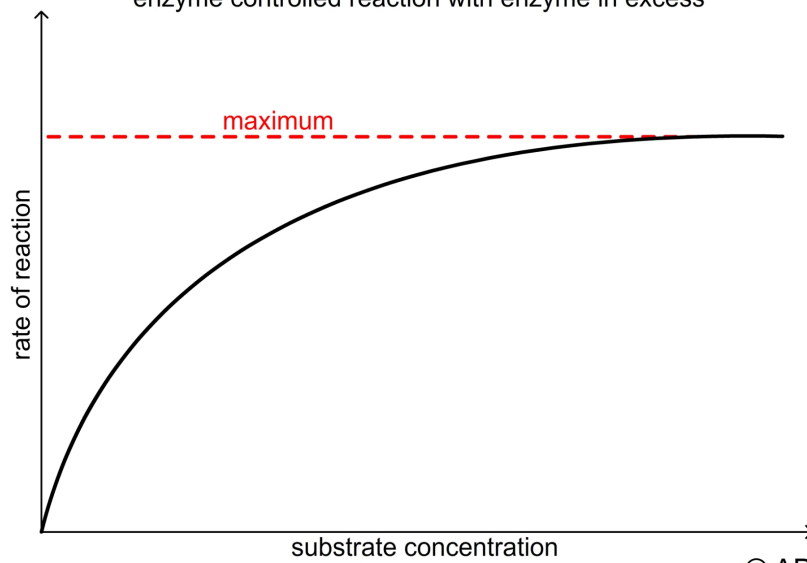
Students should be able to interpret graphs showing the effects.

NOS: Students should be able to describe the relationship between variables as shown in graphs. They should recognize that generalized sketches of relationships are examples of models in biology. Models in the form of sketch graphs can be evaluated using results from enzyme experiments.

- Explain the effects of temperature, pH and substrate concentration on enzyme structure and function with reference to collision theory, temporary and permanent denaturation.
Increasing temp to get to optimal temp, increases the movement of enzyme and substrate, meaning more collisions
Once temp passes optimal temp, the enzymes vibrate more making the bond more easily breakable, therefore denatured
Increasing pH to get to optimal pH, increases enzyme activity increases, meaning collisions between enzyme and substrate
Once pH passes optimal pH level, the enzyme is denatured and can no longer bind with substrate, permanently denature
Increasing substrate concentration means more chance of collision making reaction rate higher
- Draw and interpret graphs showing the effects of temperature, pH and substrate concentration of the activity of enzymes.



A graph to show effect of substrate concentration on rate of an enzyme controlled reaction with enzyme in excess



C1.1.9- Measurements in enzyme-catalyzed reactions. *Students should determine reaction rates through experimentation and using secondary data.*

- Identify the manipulated (independent), responding (dependent) and controlled variation in experiments of enzyme catalyzed reactions.

Independent variable: temperature, pH, substrate concentration

Dependant variable: quantity that is measured to calculate the reaction rate

controlled variables: all factors that can affect the enzyme activity (if using substrate concentration, then temp and pH need to be controlled, vise versa)

- State the unit for enzyme reaction rate.

Amount of reaction per unit of time

- State two methods for determining the rate of enzyme reaction rates.

Measure products increasing

Measure reactants decreasing

- Describe three investigative techniques for measuring the activity of an example enzyme.
 1. Measuring oxygen production
 2. Measuring mass loss
 3. Measuring pH change
 - (4. Colorimetry)

C1.1.10- Effect of enzymes on activation energy. *Students should appreciate that energy is required to break bonds within the substrate and that there is an energy yield when bonds are made to form the products of an enzyme- catalyzed reaction. Students should be able to interpret graphs showing this effect.*

- Define activation energy.

Total energy needed to start a reaction

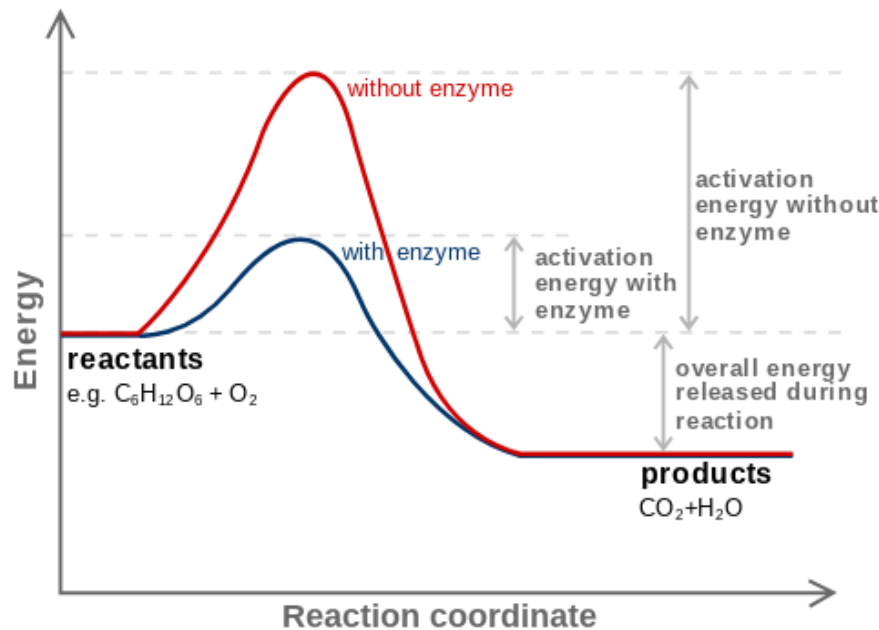
- State that activation energy is used to break or weaken bonds in the substrate.

- Explain the role of enzymes in lowering the activation energy of a reaction.

With enzyme, takes up less activation energy to get reaction to happen (by, weakening/breaking bonds, forming new bonds, reorienting atoms in the substrate)

Without enzyme, it takes more activation energy to get reaction to happen

- Interpret graphs showing the effect of lowering the activation energy by enzymes.



C1.1.11- Intracellular and extracellular enzyme-catalyzed reactions. *Include glycolysis and the Krebs cycle as intracellular examples and chemical digestion in the gut as an extracellular example.*

- Compare the location of synthesis of enzymes used within and outside of a cell.

Extracellular enzymes (exoenzymes) synthesized by ribosomes

Catalysis the breakdown of larger macromolecules

Intracellular enzymes, synthesized by ribosomes in cytoplasm

They catalyse metabolic reactions pathways

- State an example of an intracellular metabolic reaction and an extracellular metabolic reaction.

Chemical digestion in guts extracellular

Glycolysis + kreb cycle intracellular

C1.1.12- Generation of heat energy by the reactions of metabolism. *Include the idea that heat generation is inevitable because metabolic reactions are not 100% efficient in energy transfer. Mammals, birds and some other animals depend on this heat production for maintenance of constant body temperature.*

- Outline the generation of heat energy by the reactions of metabolism.
Generation of heat energy inevitable bc metabolic reactions are not 100% efficient in energy transfer
- Describe how birds and mammals maintain a body temperature greater than that of their environment.
Birds and humans use generation of heat to keep warm, body heat, warm blooded, high metabolic rate (eat more)

Birds and mammals maintain body temp using heat generated by high metabolic rates and insulation

- Outline an example of maintaining temperature homeostasis using heat generated by reactions of metabolism.
When exercising, human body produces sweat and uses evaporative cooling to dissipate excess metabolic heat

C1.1.13- Cyclical and linear pathways in metabolism. *Use glycolysis, the Krebs cycle and the Calvin cycle as examples.*

- State the reason for metabolic pathways.
Each step uses a different enzyme, which changes something in the substrate, getting to the end product
- Contrast linear metabolic pathways with cyclical reaction pathways.
Linear is a chain of steps from a reactant to a product, final product not used
Cyclical has reactants and products at the end which the reactants can be reused for the cycle, intermediates can be regenerated
- State an example of a linear metabolic pathway and a cyclic metabolic pathway.
Linear: digestion of starch
Cyclic: krebs cycle and calvin cycle

C1.1.14- Allosteric sites and non-competitive inhibition. *Students should appreciate that only specific substances can bind to an allosteric site. Binding causes interactions within an enzyme that lead to conformational changes, which alter the active site enough to prevent catalysis. Binding is reversible.*

- Outline the structure and function of an allosteric site.
Allosteric site: not the active site in the enzyme
Function: where inhibitor can bind, causing the enzyme to change shape, allow the activity of enzyme to be regulated = can prevent other substance to not bind into active site or catalyze a reaction

- Define enzyme inhibitor
Substrate that either inhibits the active site or the allosteric site, reducing enzyme activity
- Describe mechanism of action of non-competitive enzyme inhibitors.
Non-competitive enzyme inhibitors do not compete for the active site, results in enzyme changing shape, rendering the active site nonfunctional

C1.1.15- Competitive inhibition as a consequence of an inhibitor binding reversibly to an active site. *Use statins as an example of competitive inhibitors. Include the difference between competitive and non- competitive inhibition in the interactions between substrate and inhibitor and therefore in the effect of substrate concentration.*

- Describe mechanism of action of competitive enzyme inhibitors.
competitive inhibitor occupies the active site and blocks substrate
- Outline the function of statins as an example of a competitive inhibitor.
Statins, lower cholesterol by binding to active site and blocking metabolic chain, making cholesterol reduced
- Explain why the rate of reaction with increasing substrate concentration is lower with a non-competitive inhibitor compared to a competitive inhibitor.
At high substrate concentration, competitive inhibition can be overcome because substrate outcompetes the inhibitor, not possible with non-competitive inhibition

C1.1.16- Regulation of metabolic pathways by feedback inhibition. *Use the pathway that produces isoleucine as an example of an end product acting as an inhibitor.*

- Outline the mechanism and benefit of feedback inhibition.
The end product deactivates the first enzyme (inhibits)
Benefit: prevents production of product being made when high levels of product occur
- Illustrate end-product inhibition of the threonine to isoleucine metabolic pathway.
Isoleucine (end product) goes to first enzyme, binds to allosteric site, preventing more production of threonine (which is the starting substrate)

C1.1.17- Mechanism-based inhibition as a consequence of chemical changes to the active site caused by the irreversible binding of an inhibitor. *Use penicillin as an example. Include the change to transpeptidases that confers resistance to penicillin.*

- Compare reversible and irreversible enzyme inhibition.
Reversible: enzyme can go back to original shape
Irreversible: enzyme cannot go back to original shape,
- Outline the cause and consequence of mechanism-based inhibition

Inhibitor permanently binds to active site bc of covalent bond, makes enzyme never work as a catalyst again

Causes harm to organism, can kill if function of inhibited enzyme is vital and is permanently not working

- Illustrate mechanism-based inhibition using penicillin as an example.
inhibits cell wall enzyme, making enzyme not build cell walls